

## Recognizer

A recognizer is a program that takes as input a string  $X$  and answers *yes* if  $X$  is a sentence of the language and *no* otherwise.

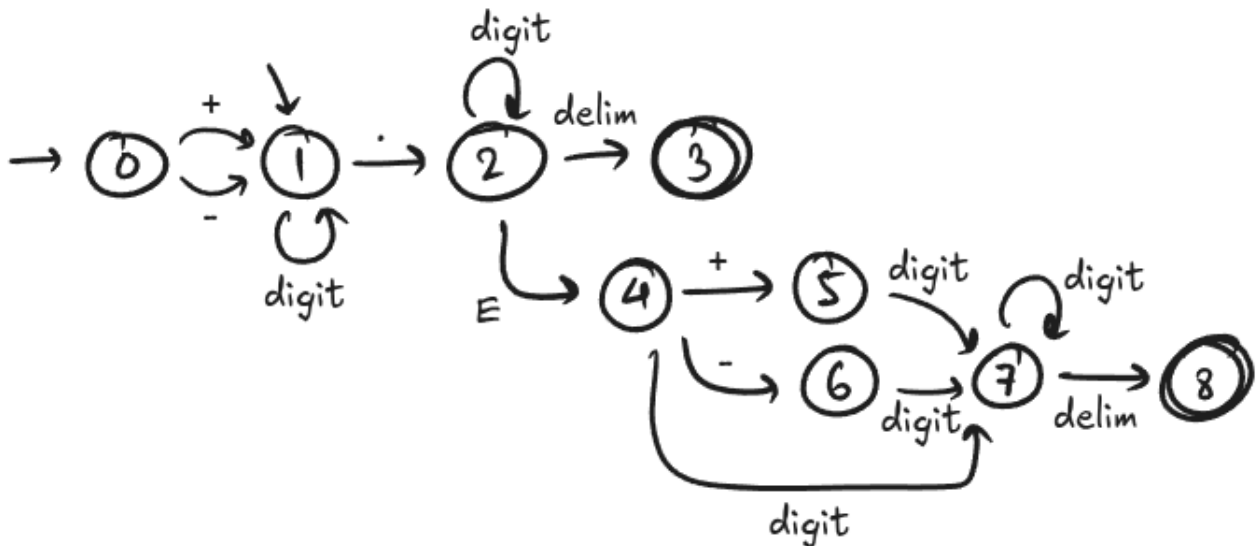


Figure 1: Recognizer for floating numbers

## Conversion of NFA to DFA using Subset Construction Algorithm

**Input:** An NFA,  $N$ .

**Output:** A DFA,  $D$  accepting the same language.

### Method:

- This algorithm constructs a transition table  $Dtran$  for  $D$ .
- Each DFA state is a state of NFA state and we construct  $Dtran$  which will simulate in parallel, all possible moves  $N$  can make on a given input string.

### Notation:

- $s$  is a NFA state
- $T$  is a set of NFA states.

### Operations:

- $\epsilon$ -closure( $s$ ): Set of NFA states reachable from NFA state  $s$  on  $\epsilon$  transition alone.
- $\epsilon$ -closure( $T$ ): Set of NFA states reachable from some NFA states in  $T$  on  $\epsilon$  transition alone. It is equal to the union of  $\epsilon$ -closure( $s$ ) for each state  $s$  in  $T$ .
- move( $T, a$ ): Set of NFA states to which there is a transition on input symbol  $a$  from some NFA state  $s$  in  $T$ .

### Algorithm:

Initially  $\epsilon$ -closure( $s_0$ ) is the only state in  $D$ -states and it is unmarked.

while there is an unmarked state  $T$  in  $D$ -states do

begin

mark  $T$

for each input symbol  $a$  do

begin

$U = \epsilon$ -closure(move( $T, a$ ))

if  $U$  is not in  $D$ -states then

add  $U$  as an unmarked state to  $D$ -states

$Dtran[T, a] = U$

end

end

To find  $\epsilon$ -closure( $T$ ):

push all states of  $T$  into *stack*

initialize  $\epsilon$ -closure( $T$ ) to  $T$

while (*stack* is not empty) {

pop  $t$  off the *stack*

for each  $\epsilon$ -transition  $u$  from  $t$

if  $u$  is not in  $\epsilon$ -closure( $T$ )

push  $u$  into *stack*

add  $u$  to  $\epsilon$ -closure( $T$ )

}

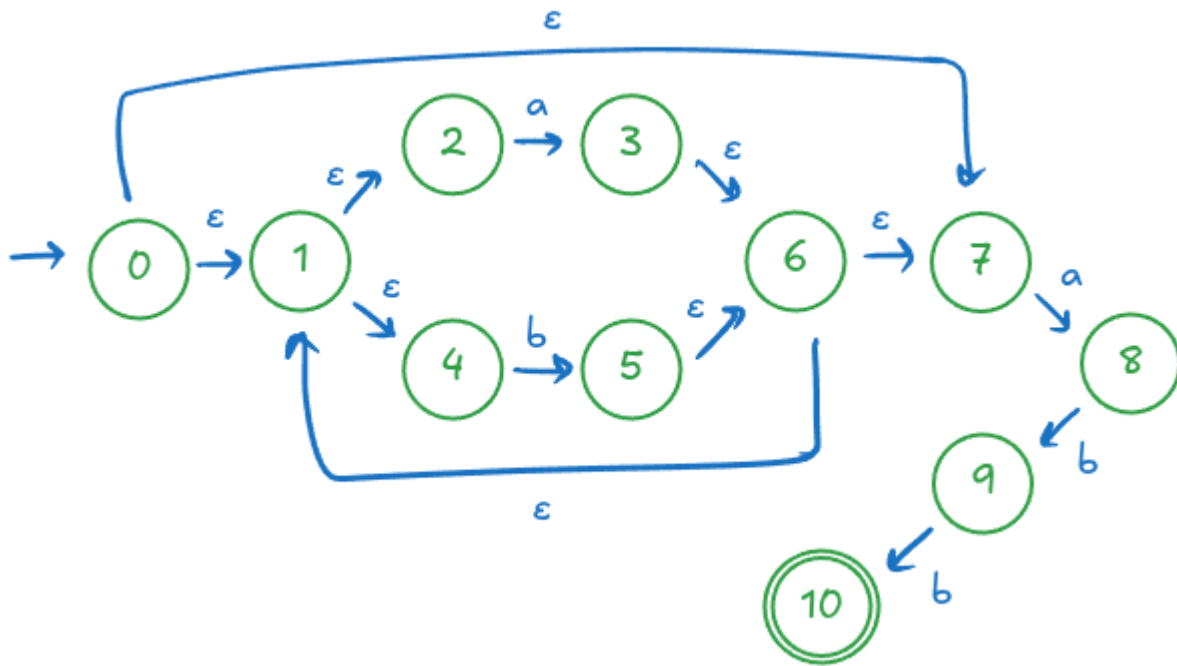


Figure 2: NFA accepting  $(a/b)^*abb$

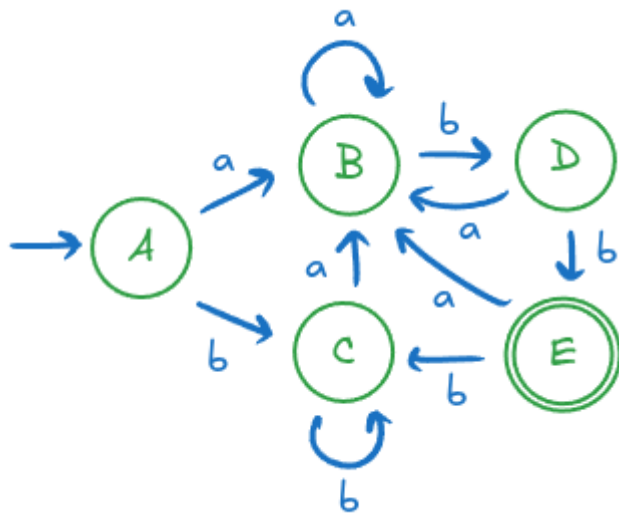


Figure 3: DFA constructed using subset construction method

| Operation                                                                                 | D-states                       | a | b |
|-------------------------------------------------------------------------------------------|--------------------------------|---|---|
| $\epsilon\text{-closure}(0)$                                                              | $\{0, 1, 2, 4, 7\} = A$        | B | c |
| $\epsilon\text{-closure}(\text{move}(A, a))$<br>$= \epsilon\text{-closure}(\{3, 8\})$     | $\{1, 2, 3, 4, 6, 7, 8\} = B$  | B | D |
| $\epsilon\text{-closure}(\text{move}(A, b))$<br>$= \epsilon\text{-closure}(\{5\})$        | $\{1, 2, 4, 5, 6, 7\} = C$     | B | c |
| $\epsilon\text{-closure}(\text{move}(B, a))$<br>$= \epsilon\text{-closure}(\{3, 8\}) = B$ |                                |   |   |
| $\epsilon\text{-closure}(\text{move}(B, b))$<br>$= \epsilon\text{-closure}(\{5, 9\})$     | $\{1, 2, 4, 5, 6, 7, 9\} = D$  | B | E |
| $\epsilon\text{-closure}(\text{move}(C, a))$<br>$= \epsilon\text{-closure}(\{3, 8\}) = B$ |                                |   |   |
| $\epsilon\text{-closure}(\text{move}(C, b))$<br>$= \epsilon\text{-closure}(\{5\}) = C$    |                                |   |   |
| $\epsilon\text{-closure}(\text{move}(D, a))$<br>$= \epsilon\text{-closure}(\{3, 8\}) = B$ |                                |   |   |
| $\epsilon\text{-closure}(\text{move}(D, b))$<br>$= \epsilon\text{-closure}(\{5, 10\})$    | $\{1, 2, 4, 5, 6, 7, 10\} = E$ | B | c |
| $\epsilon\text{-closure}(\text{move}(E, a))$<br>$= \epsilon\text{-closure}(\{3, 8\}) = B$ |                                |   |   |
| $\epsilon\text{-closure}(\text{move}(E, b))$<br>$= \epsilon\text{-closure}(\{5\}) = C$    |                                |   |   |

Table 1: Dtran table used to convert DFA in Figure 3